Special Relativity Notes

Covariant/Contravariant Expansions and the Gradient Operator in Curvilinear Coordinates

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1 Expansion in Terms of Base Vectors $\vec{\epsilon}_i$

$$\vec{A} = A^i \vec{\epsilon}_i = A_i \vec{\epsilon}^i$$

Exercise: Show that they are equivalent.

Cartesian coordinates

$$\vec{A} = A^1 \hat{e}_1 + A^2 \hat{e}_2 + A^3 \hat{e}_3$$

Proof of equivalence

Start from $\vec{A} = A^i \vec{\epsilon}_i$ and insert $\delta^j_i = g_{ik} g^{kj}$ (the defining property of the inverse metric):

$$\begin{aligned} \vec{A} = A^i \vec{\epsilon}_i &= A^i \big(g_{ik} g^{kj}\big) \vec{\epsilon}_j \\ &= \big(A^i g_{ik}\big) \big(g^{kj} \vec{\epsilon}_j\big) \\ &= A_k \big(g^{kj} \vec{\epsilon}_j\big) \qquad (\because A_k = g_{ik} A^i, \text{ lowering the index}) \\ &= A_k \, \vec{\epsilon}^k \qquad (\because \vec{\epsilon}^k = g^{kj} \vec{\epsilon}_j, \text{ raising the basis vector}) \end{aligned}$$

$$\boxed{\vec{A} = A^i \vec{\epsilon}_i = A_i \vec{\epsilon}^i}$$

Relabelling the dummy index $k \to i$ confirms the two expansions are equivalent.

2 Gradient ($\vec{\nabla}$)

Let $\Psi(q^i)$, with $q^i = q^i(x, y, z)$.

$$\vec{\nabla}\Psi = \frac{\partial \Psi}{\partial q^i} \left(\frac{\partial q^i}{\partial x} \hat{e}_x + \frac{\partial q^i}{\partial y} \hat{e}_y + \frac{\partial q^i}{\partial z} \hat{e}_z \right)$$

Recognising the bracket as the dual basis vector $\vec{\epsilon}^i$:

$$\vec{\nabla}\Psi = \frac{\partial \Psi}{\partial q^i}\vec{\epsilon}^{\,i}$$

So the **covariant** components of the gradient are read off directly:

$$(\vec{\nabla}\Psi)_i = \frac{\partial \Psi}{\partial q^i}$$

and the **contravariant** components follow by raising the index with the metric:

$$(\vec{\nabla}\Psi)^i = g^{ij}(\vec{\nabla}\Psi)_j$$

3 Worked Example: Gradient in Spherical Coordinates (r, θ, ϕ)

$$\vec{\nabla} = \frac{\partial}{\partial r}\vec{\epsilon}^{\,r} + \frac{\partial}{\partial \theta}\vec{\epsilon}^{\,\theta} + \frac{\partial}{\partial \phi}\vec{\epsilon}^{\,\phi} = \hat{r}\,\frac{\partial}{\partial r} + \frac{1}{r}\hat{\theta}\,\frac{\partial}{\partial \theta} + \frac{1}{r\sin\theta}\hat{\phi}\,\frac{\partial}{\partial \phi}$$

With $x = r\sin\theta\cos\phi$, $y = r\sin\theta\sin\phi$, $z = r\cos\theta$, $r = \sqrt{x^2+y^2+z^2}$ (as derived earlier):

$$\vec{\epsilon}^{\,r} = \frac{\partial r}{\partial x}\hat{x} + \frac{\partial r}{\partial y}\hat{y} + \frac{\partial r}{\partial z}\hat{z} = \frac{x}{r}\hat{x} + \frac{y}{r}\hat{y} + \frac{z}{r}\hat{z} = \frac{1}{r}\vec{r} = \hat{r}$$

$$\vec{\epsilon}^{\,\theta} = \frac{1}{r}\cos\theta\cos\phi\,\hat{x} + \frac{1}{r}\cos\theta\sin\phi\,\hat{y} - \frac{1}{r}\sin\theta\,\hat{z}$$

$$\vec{\epsilon}^{\,\phi} = -\frac{1}{r\sin\theta}\sin\phi\,\hat{x} + \frac{1}{r\sin\theta}\cos\phi\,\hat{y}$$

Magnitudes:

$$|\vec{\epsilon}^{\,r}| = 1, \qquad |\vec{\epsilon}^{\,\theta}| = \frac{1}{r}, \qquad |\vec{\epsilon}^{\,\phi}| = \frac{1}{r\sin\theta}$$

So the dual basis vectors relate to the unit vectors by

$$\vec{\epsilon}^{\,r} = \hat{r}, \qquad \vec{\epsilon}^{\,\theta} = \frac{\hat{\theta}}{r} \quad \left(\hat{\theta} = \frac{\vec{\epsilon}^{\,\theta}}{|\vec{\epsilon}^{\,\theta}|}\right), \qquad \vec{\epsilon}^{\,\phi} = \frac{\hat{\phi}}{r\sin\theta} \quad \left(\hat{\phi} = \frac{\vec{\epsilon}^{\,\phi}}{|\vec{\epsilon}^{\,\phi}|}\right)$$

Substituting back into $\vec{\nabla}\Psi = \dfrac{\partial \Psi}{\partial q^i}\vec{\epsilon}^{\,i}$:

$$\boxed{\vec{\nabla}\Psi = \hat{r}\,\frac{\partial \Psi}{\partial r} + \frac{1}{r}\hat{\theta}\,\frac{\partial \Psi}{\partial \theta} + \frac{1}{r\sin\theta}\hat{\phi}\,\frac{\partial \Psi}{\partial \phi}}$$

the familiar gradient operator in spherical coordinates.